Data Structures and Algorithms - Assignment 1

1. Write all possibilites to check palindrome(number, array, string, linked list, tree) on paper. Also do time and space complexity analysis of all and compare.
2. Write a linear search algorithm to return index of last occurance of key.
3. Implement linear search algorithm to find the nth occurence of the given element. If nth occurrence is not found, return -1.
4. Implement binary search algorithm if array is sorted in descending order
5. Write a recursive method to find nth term of fibbonacci series.
6. https://leetcode.com/problems/remove-all-adjacent-duplicates-in-string
7. Leetcode problem : https://leetcode.com/problems/search-insert-position/description/
8. find the first non-repeating element: Input: { 1, 2, 3, -1, 2, 1, 0, 4, -1, 7, 8 } Ouput: 3

Optional

9. Find GCD of two given numbers using recursion.
10. Solve Towers of Hanoi problem using recursion.
11. https://leetcode.com/problems/contains-duplicate

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